Assignment 2 Week 1

$x = [p_x, p_y, v_x, v_y]^T \qquad u = [a_x, a_y]^T$

$\dot{x} = Ax + Bu \qquad (\dot{p} = v, \ \dot{v} = a)$

$$\begin{bmatrix} \dot{p}_x \\ \dot{p}_y \\ \dot{v}_x \\ \dot{v}_y \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} p_x \\ p_y \\ v_x \\ v_y \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} a_x \\ a_y \end{bmatrix}$$

$A^2 = 0 \longrightarrow A$ is nilpotent

RK4 method:

$$x_{k+1} = x_k + \frac{h}{6}(K_1 + 2K_2 + K_3 + K_4)$$

where, (let $f(x,u) = Ax + Bu$)

h is the time step

$K_1 = f(x_k, u_k)$

$K_2 = f(x_k + \frac{h}{2}K_1, u_k)$

$K_3 = f(x_k + \frac{h}{2}K_2, u_k)$

$K_4 = f(x_k + hK_3, u_k)$

$\therefore \ K_1 = Ax_k + Bu_k$

$K_2 = A(x_k + \frac{h}{2}(Ax_k + Bu_k)) + Bu_k$

$\Downarrow$

$K_2 = Ax_k + \frac{h}{2}\underbrace{\cancel{A^2}}_{0}x_k + \frac{h}{2}AB\,u_k + Bu_k$

$= Ax_k + \frac{h}{2}ABu_k + Bu_k$

$K_3 = A(x_k + \frac{h}{2}(Ax_k + \frac{h}{2}ABu_k + Bu_k)) + Bu_k$

$= Ax_k + \frac{h}{2}\underbrace{\cancel{A^2}}_{0}x_k + \frac{h^2}{4}\underbrace{\cancel{A^2}}_{0}Bu_k + \frac{h}{2}ABu_k + Bu_k$

$K_3 = Ax_k + \frac{h}{2}ABu_k + Bu_k = K_2$

$K_4 = A(x_k + h(Ax_k + \frac{h}{2}ABu_k + Bu_k)) + Bu_k$

$= Ax_k + h\underbrace{\cancel{A^2}}_{0}x_k + \frac{h^2}{2}\underbrace{\cancel{A^2}}_{0}Bu_k + hABu_k + Bu_k$

$= Ax_k + hABu_k + Bu_k$

$\therefore \ x_{k+1} = x_k + \frac{h}{6}(6Ax_k + 3hABu_k + 6Bu_k)$

$\therefore \ x_{k+1} = x_k + hAx_k + \frac{h^2}{2}ABu_k + hBu_k$

$\therefore \ x_{k+1} = (I + hA)x_k + \left(\frac{h^2}{2}AB + hB\right)u_k$

$$AB = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$\therefore \ x_{k+1} = \begin{bmatrix} 1 & 0 & h & 0 \\ 0 & 1 & 0 & h \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} x_k + \begin{bmatrix} h^2/2 & 0 \\ 0 & h^2/2 \\ h & 0 \\ 0 & h \end{bmatrix} u_k$$

$$\therefore \begin{bmatrix} p_x \\ p_y \\ v_x \\ v_y \end{bmatrix}_{k+1} = \begin{bmatrix} 1 & 0 & \Delta t & 0 \\ 0 & 1 & 0 & \Delta t \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} p_x \\ p_y \\ v_x \\ v_y \end{bmatrix}_k + \begin{bmatrix} \Delta t^2/2 & 0 \\ 0 & \Delta t^2/2 \\ \Delta t & 0 \\ 0 & \Delta t \end{bmatrix} \begin{bmatrix} a_x \\ a_y \end{bmatrix}_k$$

To handle the "non-convexity" of obstacles, we use the separating hyperplanes approach, defining the obstacle by the physical bounds $[x_{min}, x_{max}]$ and $[y_{min}, y_{max}]$, such that at each time step, the drone avoids collision if at least one of these constraints are satisfied:

$p_{x,k} \leq x_{min}$

$p_{x,k} \geq x_{max}$

$p_{y,k} \leq y_{min}$

$p_{y,k} \geq y_{max}$

The 'at-least' condition can be formulated as

$p_{x,k} - x_{min} \leq M(1 - z_{1,k})$

$x_{max} - p_{x,k} \leq M(1 - z_{2,k})$

$p_{y,k} - y_{min} \leq M(1 - z_{3,k})$

$y_{max} - p_{y,k} \leq M(1 - z_{4,k})$

$$\sum_{i=1}^{4} z_{i,k} \geq 1$$

where,

$z_{i,k} \in \{0, 1\}$ for $i = 1, 2, 3, 4$

M is a sufficiently large positive constant